

1. Create Database

Purpose

Creates a new database to store supermarket data.

Query:

```
CREATE DATABASE supermarket_db;
```

2. Select Database

Purpose

Selects the database for performing SQL operations.

Query:

```
USE supermarket_db;
```

3. Create Table

Purpose

Creates a table to store product information.

Query:

```
CREATE TABLE stock_management (  
    p_id INT,  
    p_name VARCHAR(30),  
    p_price INT,  
    p_stock INT  
);
```

4. INSERT INTO

Purpose

Adds records to the table.

Query:

```
INSERT INTO stock_management VALUES (110,'paneer',25,20);
```

Other inserted products:

- Paneer
- Milk
- Curd

- Pen
- Paper
- Pencil
- Tissue

Note: Product ID **110** is inserted twice, so duplicate records exist because no PRIMARY KEY constraint was defined.

5. SELECT Statement

Purpose

Displays all records.

Query:

```
SELECT * FROM stock_management;
```

Result:

Result Grid					
Filter Rows:					
	p_id	p_name	p_price	p_stock	p_brand
▶	110	paneer	70	20	NULL
	110	paneer	70	20	NULL
	111	milk	35	21	NULL
	112	curd	50	22	NULL
	113	pen	55	20	NULL
	114	paper	70	20	NULL
	115	pencil	90	20	NULL
	116	tissue	10	19	NULL

6. ALTER TABLE

Purpose

Adds a new column to the existing table.

Query:

```
ALTER TABLE stock_management ADD p_brand VARCHAR(20);
```

7. UPDATE Statement

Purpose

Modifies existing data.

Query:

```
UPDATE stock_management SET p_price = 70 WHERE p_id = 110;
```

Result

8. SQL_SAFE_UPDATES

Purpose

Disables MySQL Safe Update mode.

Query:

```
SET SQL_SAFE_UPDATES = 0;
```

Why?

Allows UPDATE and DELETE statements even when MySQL considers them unsafe.

9. WHERE Clause

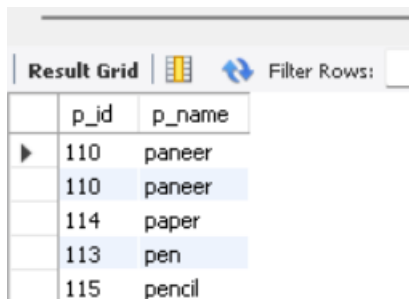
Purpose

Filters rows based on a condition.

Query:

```
SELECT p_id, p_name FROM stock_management WHERE p_price > 50;
```

Result



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with two columns: 'p_id' and 'p_name'. There are five rows of data. The first two rows have 'p_id' 110 and 'p_name' 'paneer'. The third row has 'p_id' 114 and 'p_name' 'paper'. The fourth row has 'p_id' 113 and 'p_name' 'pen'. The fifth row has 'p_id' 115 and 'p_name' 'pencil'. The first two rows are highlighted in blue.

	p_id	p_name
▶	110	paneer
	110	paneer
	114	paper
	113	pen
	115	pencil

10. ORDER BY

Purpose

Sorts records.

Ascending Order

```
SELECT p_id, p_name FROM stock_management WHERE p_price > 50 ORDER BY p_name;
```

Result:

Result Grid			Filter Rows:
	p_name	total_stock	
▶	paneer	2	
	milk	1	
	curd	1	
	pen	1	
	paper	1	
	pencil	1	
	tissue	1	

Sorts product names alphabetically.

Descending Order

SELECT p_id, p_name FROM stock_management ORDER BY p_id DESC;

Displays products from highest Product ID to lowest.

Result:

Result Grid			Filter Rows:
	p_id	p_name	
▶	116	tissue	
	115	pencil	
	114	paper	
	113	pen	
	112	curd	
	111	milk	
	110	paneer	
	110	paneer	

11. GROUP BY with COUNT()

Purpose

Groups identical product names and counts the number of records.

Query:

SELECT p_name, COUNT(p_price) AS total_stock FROM stock_management GROUP BY p_name;

12. DELETE Statement

Purpose

Removes records from a table.

Query:

DELETE FROM stock_management WHERE p_id = 109;

13. GROUP BY with SUM()

Purpose

Calculates the total stock available for each product.

Query:

```
SELECT p_name, SUM(p_stock) AS total_stock FROM stock_management GROUP BY p_name;
```

Result:

Result Grid			Filter Rows:
	p_name	total_stock	
▶	paneer	40	
	milk	21	
	curd	22	
	pen	20	
	paper	20	
	pencil	20	
	tissue	19	